Genetic Algorithm Search for Stable Vacua in Type IIB String Theory: A Computational Approach

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Abstract. The exploration of high-dimensional, non-linear potential landscapes is a central computational challenge. In fields such as string theory, the search for stable vacuum configurations reduces to a global optimisation problem on a rugged objective function where classic gradient-based methods are inadequate. Previous work demonstrated the potential of nature-inspired algorithms in this context.
In this work, a simple genetic algorithm (GA) is tailored to the optimisation of a type IIB string theory potential with torsion, extending the analysis of a previous work done with Simulate Annealing (which found 203 stable dS and 529 stable AdS). A penalty-based objective enforces critical point conditions, stability and a positive potential. Using uniform bounds for the five flux coefficients and 100 independent runs, the GA explored the landscape extensively, producing 11,979 critical points. The behaviour of the potential under the $\widehat{D5}$ uplifting term was analysed, revealing type transitions, modulus-specific effects and consistent negativity of $A_3\mathcal{N}_3$ without explicit constraints. The results confirm the GA's ability to discover stable de Sitter vacua and to provide a rich dataset for further computational and theoretical study.
Finally, a grid was performed for the aim of finding better configurations for this approach. The grid analysed 4 different models over 3 different moduli configurations. The results were statistically analysed via fitness, potential value and the amount of stable points found (obtaining a stability ratio).

Keywords: Type IIB String Theory · Flux Compactifications · de Sitter Vacua · Genetic Algorithms · Global optimisation · Metaheuristics

1 Introduction

The exploration capabilities of computational nature-inspired methods have represented a significant advance in tackling non-linear models where traditional analytical or gradient-based techniques struggle [15, 11]. Among these, genetic algorithms (GA) stand out as a robust and widely applicable metaheuristic. Inspired by natural selection, GAs operate on populations of candidate solutions, evolving them through selection, crossover, and mutation. This population-based approach enables effective exploration of high-dimensional, multimodal, and ill-behaved solution spaces without requiring convexity or differentiability assumptions [14, 6]. Consequently, GAs have become a tool of choice for complex optimisation problems in engineering and science.

In string theory, the exploration of the landscape—the set of consistent vacuum configurations—poses a computational challenge that exemplifies such difficulties. From an optimisation standpoint, the problem reduces to finding minima of a high-dimensional potential energy function. The resulting landscape is rugged: it contains numerous local optima, strongly non-convex regions, and numerical instabilities that render deterministic methods ineffective [10, 5]. As the dimensionality and complexity of these models increase, the need for robust, scalable computational tools becomes critical [12].

Recent work has demonstrated the potential of genetic algorithms within this domain. [1] showed that GAs are orders of magnitude more efficient than random sampling for locating viable string vacua. Hybrid approaches combining GAs with neural networks have enabled the exploration of exponentially larger spaces, yielding thousands of flux configurations with critical points [2]. Beyond efficiency, GAs have been shown to discover entirely new configurations not present in exhaustive catalogs [9], highlighting their capacity for exploratory discovery.

A particularly relevant study by [4] employed simulated annealing—a single metaheuristic—to identify over 170 metastable de Sitter vacua in type IIB compactifications with torsion. While that work demonstrated the feasibility of the approach, it left open the question of whether a population-based method like GA could achieve better performance in terms of both solution quality and the number of stable vacua discovered. The No Free Lunch theorem [13] reminds us that no single optimisation algorithm is universally superior; the relative performance depends on the specific properties of the objective function and search space. Therefore, applying a GA to this problem and comparing its results against previously reported ones provides valuable insight into the suitability of evolutionary strategies for exploring the string landscape.

In this work we address this question from a computational perspective. We implement a genetic algorithm tailored to the type IIB potential model used in [4]. To ensure a fair and reproducible baseline, we have developed a complete symbolic implementation of the potential, its gradient, and its Hessian in Python, validated against the original Mathematica code. The GA is evaluated using the same initial configuration database and performance metrics employed in the reference study, including convergence speed, solution quality, and the number of new stable vacua discovered.

The contributions of this paper are twofold. First, we provide a systematic assessment of a genetic algorithm applied to a concrete high-dimensional optimisation problem arising from string theory, offering a comparative perspective against a previously published simulated-annealing baseline. By bridging computational intelligence and theoretical physics, this work aims to establish a robust methodological foundation for tackling complex optimisation problems across disciplines.

2 Related Work

The application of computational intelligence to string theory has grown steadily over the past decade, with metaheuristics playing an increasingly central role in navigating the enormous space of possible vacuum configurations. In the work described in [1], Abel and Rizos demonstrated that genetic algorithms (GAs) can efficiently explore the string landscape in free fermionic models. Despite a search space of astronomical size—estimated at 2^{51} possibilities—the GA was shown to be orders of magnitude more efficient than random sampling at locating vacua with specific phenomenological properties. This result was significant because it revealed that the landscape's structure, though complex, is amenable to evolutionary search, where selection and recombination guide the population toward promising regions.

Building on this, Bizet and collaborators directed the research of a hybrid approach in [2], combining a GA with a neural network to explore type IIB flux compactifications. Their method identified tens of thousands of flux configurations with critical points, dramatically reducing the computational cost compared to brute-force enumeration. The synergy between the GA's global exploration and the neural network's ability to approximate the potential landscape illustrated how machine learning can amplify the power of metaheuristics in high-dimensional physics problems.

Beyond efficiency, metaheuristics have also proven capable of discovering entirely new configurations. He, Jejjala, and Silva observed in [9] that simulated annealing—a classic stochastic optimisation technique—can generate brane tilings, geometric constructions central to certain string compactifications. Their search yielded a previously unknown tiling with 26 quantum fields, a configuration absent from exhaustive catalogs compiled by traditional methods. This work underscored that metaheuristics are not merely optimizers but can serve as discovery engines, uncovering novel structures that human intuition or systematic enumeration might miss.

Most directly relevant to the present study is the work described in [4], where Damian and Loaiza-Brito focused on metastable de Sitter vacua in type IIB compactifications with torsion. The authors implemented simulated annealing combined with a conjugate-gradient local refinement to identify over 203 stable de Sitter vacuum and 170 stable Anti de Sitter vacuum configurations. Their approach provided a concrete baseline for the potential model we adopt, and their code—originally written in Mathematica—offered a detailed reference for the symbolic form of the potential, its gradient, and its Hessian. However, their study relied on a single stochastic method, leaving open the question of whether other metaheuristic families, particularly population-based algorithms like GAs, could achieve better performance in terms of convergence speed, solution quality, or the number of vacua discovered.

Other recent efforts have explored complementary computational strategies. Cole and collaborators directed the research described in [3], using both reinforcement learning and GAs to probe the structure of flux vacua, revealing previously unidentified symmetries and emphasizing the importance of combining multiple search methods to reduce sampling bias. In [8], Halverson and Long demonstrated that deep generative models can approximate the distribution of Kähler metrics across Calabi-Yau manifolds, capturing correlations in the landscape with far fewer samples than naive expectations. Meanwhile, Gao and Zou observed in [7] that convolutional neural networks can classify Calabi-Yau orientifolds with string vacua, achieving classification accuracies above 99.9%. Although these works are not strictly metaheuristic, they highlight the broader trend of using advanced computational tools to tackle the string landscape.

The No Free Lunch theorem, as formulated by Wolpert and Macready in [13], reminds us that no single algorithm is universally optimal; performance depends on the specific problem structure. The potential function in type IIB torsion compactifications is highly non-convex, multimodal, and numerically sensitive—characteristics that favor population-based methods capable of maintaining diversity and escaping local optima. Yet, no systematic comparison of different metaheuristic families has been performed on this model. The present work addresses this gap by implementing a GA and evaluating it against the simulated annealing baseline established in [4], thereby contributing a computational benchmark for future explorations of the string landscape.

3 Materials and Methods

3.1 optimisation Method

Problem Formulation The effective scalar potential for type IIB compactifications with torsion, including the contribution of non-BPS $\widehat{D}5$-branes, is taken from [4]:

$$\mathcal{V}_{\text{eff}} = \frac{A_{H_3} s}{\tau^3} + \frac{A_{F_3}}{s\tau^3} + \frac{A_{F_5}}{\tau^4} + \frac{A_3\mathcal{N}_3}{\tau^3} + \frac{A_{\widehat{D5}}}{s^{1/2}\tau^{5/2}}.$$

The variables are the flux coefficients $A_{H_3}, A_{F_3}, A_{F_5}, A_3\mathcal{N}_3, A_{\widehat{D5}}$ and the moduli s, τ. A physically admissible vacuum must satisfy

$\nabla\mathcal{V}_{\text{eff}} = 0$ (minimal energy),
$\partial_i\partial_j\mathcal{V}_{\text{eff}} > 0$ (no tachyonic),
$\mathcal{V}_{\text{eff}} > 0$ (de Sitter vacuum)

Penalty-Based Objective All constraints are enforced through penalty terms, so that no candidate solution is discarded during the search. The overall fitness function $F(\mathbf{x})$ is a sum of four components:

- **Gradient penalty**: $f_{\text{grad}} = \|\nabla\mathcal{V}_{\text{eff}}\|^2$, which encourages the solution to be a critical point.
- **Positive potential penalty**: $p_{\text{pos}} = |\mathcal{V}_{\text{eff}}| - \mathcal{V}_{\text{eff}}$, which is zero when $\mathcal{V}_{\text{eff}} \geq 0$ and positive otherwise, thus favoring de Sitter vacua.
- **Trace penalty**: $p_{\text{tr}} = |\operatorname{tr}(\mathbf{H})| - \operatorname{tr}(\mathbf{H})$, where $\mathbf{H}$ is the Hessian. This term penalizes a negative trace, a necessary condition for positive definiteness.
- **Discriminant penalty**: $p_{\text{disc}} = |\Delta| - \Delta$ with $\Delta = \det(\mathbf{H}) - \frac{1}{4}(\operatorname{tr}\mathbf{H})^2$. This term penalizes a negative discriminant, which would indicate imaginary eigenvalues or a saddle point.

So the fitness function based on penalties is given by

$$F(\mathbf{x}) = f_{\text{grad}} + p_{\text{pos}} + p_{\text{tr}} + p_{\text{disc}},$$

which must be minimized.

Genetic Algorithm Configuration The optimisation uses the simple genetic algorithm (GA) from the `MEALPY` library. Where the first configurations was

- Population size: 20 individuals.
- Number of generations: 50.
- Crossover probability: 0.9.
- Mutation probability per gene: 0.4.
- Number of independent runs per search configuration: 100.

All other parameters use the library defaults: selection is tournament ($k_{\text{way}} = 0.2$), crossover is uniform, mutation is flip with multipoint mutation enabled, and elitism is applied. Each run starts from a random initial population drawn uniformly from the bounded search space.

Search Space and Bounds The moduli were fixed using values from a dataset of [4]. Thus, the search space comprises only the five flux coefficients A_{H_3}, A_{F_3}, A_{F_5}, $A_3\mathcal{N}_3$, $A_{\widehat{D5}}$. With the bound interval

$$-100 \leq A_{H_3}, A_{F_3}, A_{F_5}, A_3\mathcal{N}_3, A_{\widehat{D5}} \leq 100.$$

Physically, the combination $A_3\mathcal{N}_3$ is expected to be negative, but the optimisation treats all coefficients uniformly and imposes no sign restriction a priori.

3.2 Experiments and Results

Four independent runs of the optimisation program were executed, yielding a total of 11 979 distinct solutions. Each solution was classified according to the sign of the effective potential (de Sitter or anti-de Sitter) and the stability of the Hessian (no tachyons vs. tachyonic).

This approach was able to find 987 stable de Sitter vacua and 2 309 stable Anti de Sitter vacua. In addition, this approach found 1 345 de Sitter unstable vacua and 7 338 Anti de Sitter vacua. In terms of the swampland conjectures, this behaviour is expected due to the large amount of vacua, of which most of them tend to be anomalous.

Magnitude Comparison with the Original Database The original reference database given by [4] contains a limited set of configurations, each with fixed moduli values. In the present experiments, the moduli were kept fixed to those values while the coefficients were optimized. A direct comparison between the found coefficients and their original counterparts reveals that the optimized coefficients are generally larger in magnitude than those in the original database. Table 1 shows an illustrative comparison row from the collected data.

Table 1. Comparison of a representative vacuum solution between the present method (Found) and the original database (Original).

Parameter	Found	Original
V	-3.7285627019	-0.0001171300
A_{H_3}	13.9045540256	0.2684424662
A_{F_3}	-12.5375584819	-0.0397849749
A_{F_5}	6.3484077366	1.6359215112
$A_3\mathcal{N}_3$	-64.1172435464	-1.7877642196
$A_{\widehat{D5}}$	35.2931058421	0.6157456543
λ_1	-3.2694182084	0.0012383000
λ_2	8.9804716967	0.0060276000

The observed magnitude differences suggest that the search space may require additional constraints to align with theoretical expectations, or that the landscape contains viable vacua with larger flux values not captured by the original sampling.

Stability Changes on the Same Moduli An important behaviour given by this approach is that some optimisations found stable and unstable fluxes contributions on the same moduli, providing new behaviours of the effective potential, which allows finding new configurations by exploiting already optimised moduli, compared with [4] where the exploitation was done by conjugate gradient.

Behavioural Observations When comparing the behaviour of the non-lifted potential against the lifted potential, several distinct patterns emerged. A first category comprises points that retained their vacuum type (de Sitter or anti-de Sitter) across the two potentials. Figure 3 illustrates a representative example where the potential maintains its qualitative shape after lifting, with only minor quantitative adjustments.

Conversely, lifting induced type transitions in a significant fraction of points. Some anti-de Sitter points became de Sitter after the $\widehat{D5}$ term was introduced; a representative case is shown in Fig. 4. Likewise, a number of de Sitter points turned into anti-de Sitter vacua.

Beyond type changes, lifting affected the potential along the individual moduli in non-trivial ways. In some instances the lifting acted primarily on the dilaton s (Fig. 5), while in others the volume modulus τ dominated the response. For a subset of points, the lifting lowered the potential along the dilaton direction, the volume direction, or both, occasionally causing the potential to become negative in certain ranges or to flatten substantially, as seen in Fig. 6. Points where the potential remained qualitatively unchanged but with altered moduli behaviour are exemplified in Fig. 3.

The transition behaviour between vacuum types when the $\widehat{D5}$ term is introduced is shown in Figs. 1 and 2 for the new solutions and the original configurations, respectively. These plots highlight how the lifting term affects the vacuum classification.

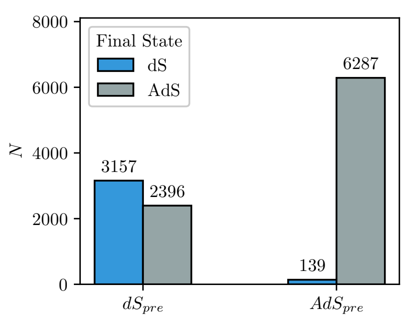

Fig. 1. Transition between vacuum types (new solutions).

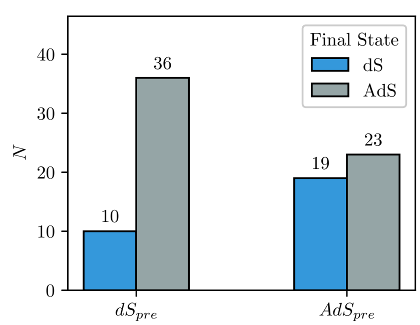

Fig. 2. Transition between vacuum types (original database).

Consistent Orientfolds Contribution (A_3N_3) Behaviour A noteworthy observation concerns the coefficient $A_3\mathcal{N}_3$. Even though no sign restriction was imposed during optimisation, the genetic algorithm consistently returned negative values for this coefficient, in agreement with the physical expectation.

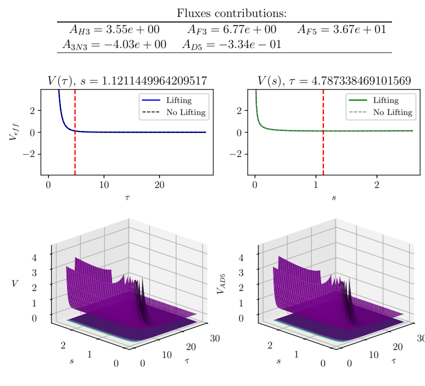

Fig. 3. A point that retained its vacuum type across the non-lifted and lifted potentials.

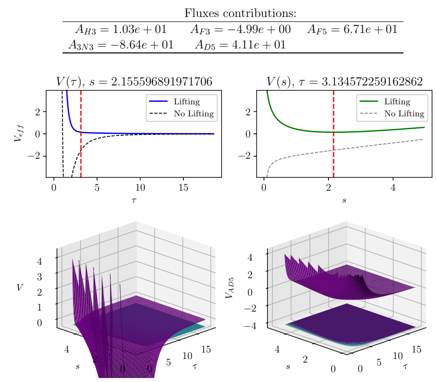

Fig. 4. Transition from anti-de Sitter to de Sitter after lifting.

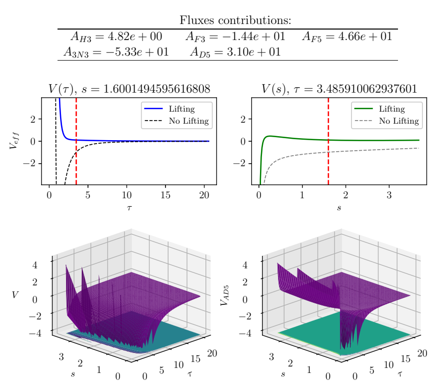

Fig. 5. Lifting acting primarily on the dilaton s.

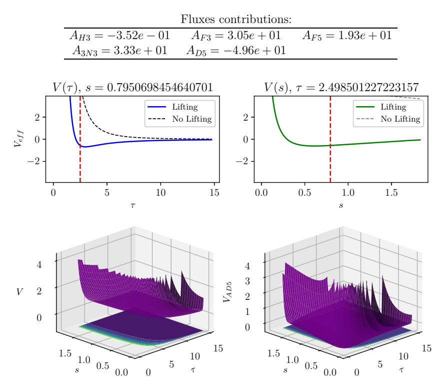

Fig. 6. Lowering of the potential along the moduli directions after lifting, leading to negative regions or flattening.

3.3 Grid experiment

For further analysis an experiment was done regaring a grid of `Basic GA` configurations. Originally the grid was planned for around 800 different configurations.

- **Epoch:** [50, 100, 200, 300, 400]
- **Population Size (pop_size):** [20, 60, 100, 140, 180, 220, 260, 300]
- **Crossover Probability (pc):** [0.1, 0.3, 0.5, 0.7, 0.9]
- **Mutation Probability (pm):** [0.2, 0.4, 0.6, 0.8]

For testing different models, there were 6 different instances of testing implementing different variations from which those analysed were shown in the table 2. The test was done over three different moduli configurations shown in the table 1.

Table 2. Detailed Parameters for Each Model

	Epoch	Population Size (pop_size)	Crossover Probability (pc)	Mutation Probability (pm)
Model 1	50	20	0.1	0.2
Model 2	100	60	0.9	0.8
Model 3	50	20	0.5	0.6
Model 4	50	60	0.5	0.6

Table 3. Moduli configurations (s, τ) for the three experimental runs.

Configuration	s	τ
Run 1	1.600149459561681	3.485910062937601
Run 2	2.459242111292193	3.0947272010640288
Run 3	1.053781742958659	1.4533862399266009

3.4 Results of the grid

Stability of vacua For each model and each run, the fraction of stable vacua was computed. Table 4 reports these stability ratios.

Model 3 exhibits consistently moderate-to-high stability, reaching 96 % in Run 3.

Table 4. Stability ratios (stable vacua / total points) per model and run.

Model	Run 1	Run 2	Run 3
Model 1	0.49	0.31	0.87
Model 2	0.44	0.17	0.99
Model 3	0.50	0.36	0.96
Model 4	0.43	0.25	0.98

Table 5. Shapiro–Wilk normality test results for V and fitness.

Model	Run	V		fitness	
		p-value	normal?	p-value	normal?
1	1	0.0001	No	0.1046	Yes
1	2	0.0007	No	0.0000	No
1	3	0.1623	Yes	0.0000	No
2	1	0.0000	No	0.3620	Yes
2	2	0.0036	No	0.0719	Yes
2	3	0.3750	Yes	0.3386	Yes
3	1	0.0001	No	0.5932	Yes
3	2	0.0003	No	0.0248	No
3	3	0.3651	Yes	0.0000	No
4	1	0.0000	No	0.6264	Yes
4	2	0.0042	No	0.2522	Yes
4	3	0.3894	Yes	0.0284	No

Normality assessment Because later statistical tests require normality or lack thereof, a Shapiro–Wilk test was applied to the distributions of the potential V and the **fitness** metric for every combination (Model, Run). Table 5 summarises the outcomes (p-values and decision at $\alpha = 0.05$).

Since many distributions deviate from normality, non-parametric tests were adopted for group comparisons.

Statistical comparison of models For each of the three moduli groups (Run 1, Run 2, Run 3) the four models were compared using the **fitness** metric. Because the data are not always normal, a Mann–Whitney U test was applied. To keep the family-wise error rate under control, the significance level was adjusted with a Bonferroni correction: for three pairwise comparisons per group, $\alpha_{\text{corrected}} = 0.05/3 = 0.0167$.

Within each group the model with the highest median fitness was selected as the candidate "best" model and then tested against the other three.

Visualisation of distributions Figure 8 shows violin plots comparing the potential V (upper panel) and the fitness (lower panel) across the four models, with each run shown as a different colour. The plots confirm the spread and central tendency differences observed in the statistical tests. And also a comparative histogram of stable vacua found during the grid is shown in figure 7.

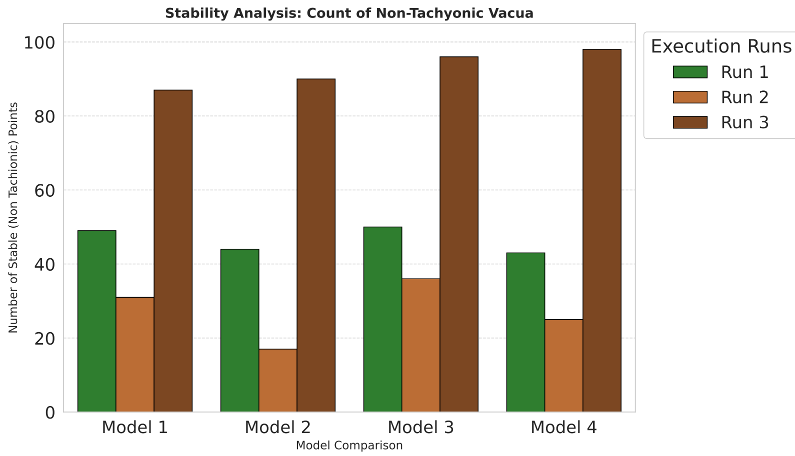

Fig. 7. Number of stable vacua found by each model in each run of the grid analysis.

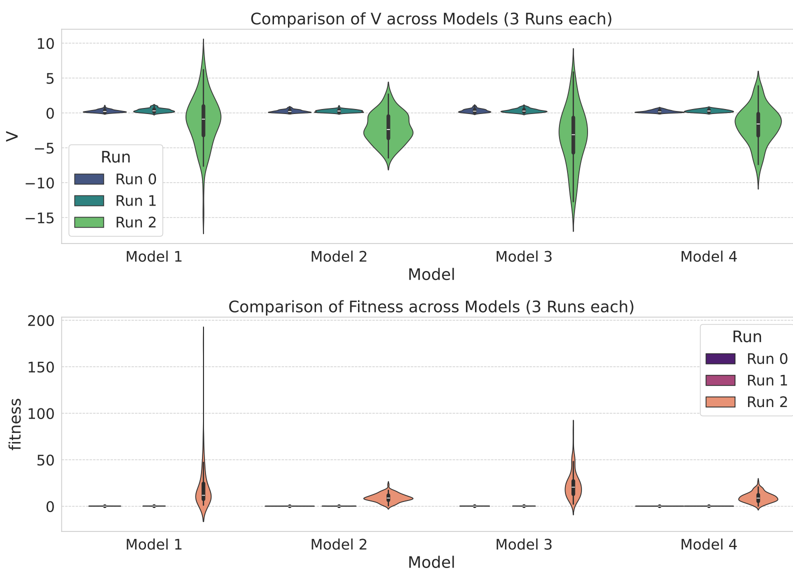

Fig. 8. Comparison of V (top) and fitness (bottom) across the four models. Each model is shown with its three independent runs.

Table 6. Median fitness values and pairwise comparison results.

Run	Best Model	Median fitness	Outcome vs others
1	3	0.1215	Statistically better than Model 1 (p=0.0044), Model 2 (p=0.0000) and Model 4 (p=0.0000). **Clear winner.**
2	3	0.1560	No statistical lead over Model 1 (p=0.0552); better than Model 2 (p=0.0000) and Model 4 (p=0.0000). **Tie with Model 1.**
3	3	20.4978	Statistically better than Model 1 (p=0.0002), Model 2 (p=0.0000) and Model 4 (p=0.0000). **Clear winner.**

3.5 Challenges and Study Limitations

Behaviour Plotting For some vacua found, the potential behaviour could not be plotted due to unknown reasons. Some suspects points towards numerical inconsistencies, asymptotic range of moduli values, or equipment limitations. The main reason to believe the points found and evaluated within the potential are actually valid is that each point evaluated in both versions of the potential exists.

Time Limitations Some models were quite extensive, and due to the complexity of the penalty function, even with GPU integration, their execution times were still in the order of days. Instabilities on the electrical service rendered these more complex models unfinished, although they seemed to perform slightly better at first sight based on some solutions noted in the logs.

3.6 Conclusions

The present work extended the analysis originally reported in [4] by replacing the simulated annealing and conjugate gradient optimisation with a simple genetic algorithm (GA). The search space was reformulated as a penalty-based optimisation over the five flux coefficients $A_{H_3}, A_{F_3}, A_{F_5}, A_3\mathcal{N}_3, A_{\widehat{D5}}$, each bounded uniformly to $[-100, 100]$. This computational approach allowed a more extensive exploration of the effective potential landscape for type IIB compactifications with torsion, yielding a total of 11 979 critical points.

The results confirmed the capability of the GA to discover stable de Sitter vacua, with counts comparable to those reported in the reference study. More importantly, the systematic comparison between the non-lifted and lifted potentials revealed several non-trivial behaviours that had not been characterised before. Among these were points that retained or changed their vacuum type under lifting, cases where the lifting acted predominantly on a single modulus (volume τ), and configurations in which the potential became significantly flatter or even developed negative regions after lifting.

Even though no sign restriction was imposed on $A_3\mathcal{N}_3$, the GA consistently returned negative values for this coefficient, in agreement with the physical expectation. This observation suggests that the penalty-based formulation did not force artificial sign choices but allowed the algorithm to converge to physically plausible regions.

Is also worth to mention that the exploitation of moduli configurations is of special interest because of the conjectures given by the swampland project.

A small fraction of the computed points exhibited numerical instabilities that prevented their inclusion in the graphical analysis. These points are mathematically valid but require higher precision or alternative algebraic handling; their study could uncover further structures of interest both from a computational and from a string-theory perspective.

On the other hand, the grid analysis found that tuning the `GA` can give better results even with a basic version of the algorithm.

Future work will extend the methodology to a wider family of nature-inspired algorithms (e.g. ant colony optimisation, estimation of distribution algorithms, weed distribution optimisation, atom search optimisation) to assess whether alternative search strategies can further improve the exploration of the landscape.

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